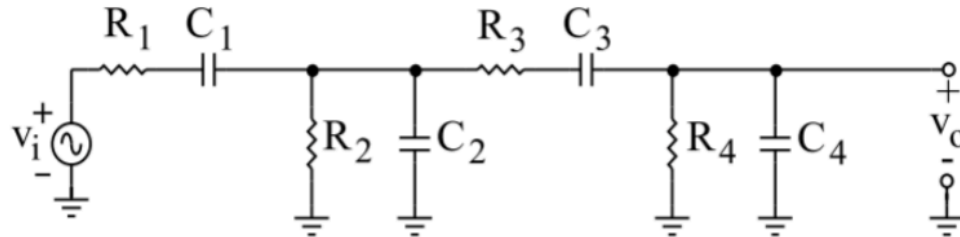


ELEC 301 - Project 1 - Edan Steen

PROBLEM 1

P1. The figure below shows a simple four-pole RC filter. With proper selection of the values of R_1 , R_2 , R_3 , R_4 , C_1 , C_2 , C_3 , and C_4 , this circuit becomes a band pass filter. We will use the method of OC and SC time constants to find the transfer function of such a filter.



Four-pole RC filter

Initial component values for the RC filter circuit:

R_1	R_2	R_3	R_4	C_1	C_2	C_3	C_4
50 Ω	500 Ω	500 Ω	500 Ω	20 μF	100 pF	500 nF	100 pF

1A. Run an AC simulation on the circuit over a frequency range from at least 3 decades below the low frequency 3-dB point to at least 3-decades above the high frequency 3-dB point. Plot the Bode plots, both magnitude and phase, for the circuit. Graphically identify and record the locations of each of the poles of the transfer function.

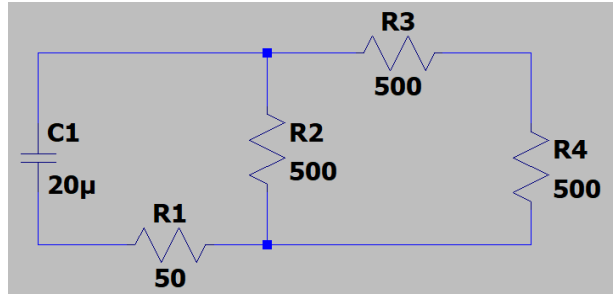
Midband gain

To run our AC simulation we have to find the transfer function for the given circuit. First we can analyze the midband gain of the circuit by shorting the capacitors C_1 and C_3 , as well as opening capacitors C_2 and C_4 . From the resulting circuit, we get $v_o = -10v_i$, giving us a midband gain of $A_M = -10$.

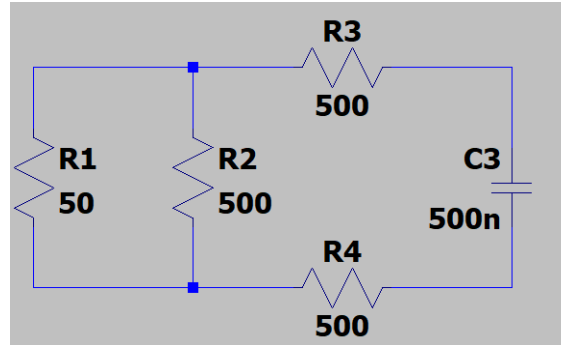
Low frequency poles

Next, we perform the SC and OC time constant tests. For the SC test, we open the high frequency capacitors C_2 and C_4 , and then get the equivalent thevenin resistances with respect to each capacitor while shorting the other. The poles are $\omega_{Ci} = 1/(R_{th}C_i)$. This gives us $\omega_{C1}=130.43$ Hz and $\omega_{C3}=1913$ Hz. The low frequency pole is found by the expression

$\omega_L = \sqrt{\omega_{C1}^2 + \omega_{C3}^2}$, which gives us $\omega_L = 1917.48$ rad/s. We can convert this value to Hz with the expression $\omega = 2\pi f$, giving us $\omega_L = 305.18$ Hz.



Circuit for finding the pole from C_1 from which we get $R_{th} = R1 + R2 // (R3 + R4)$

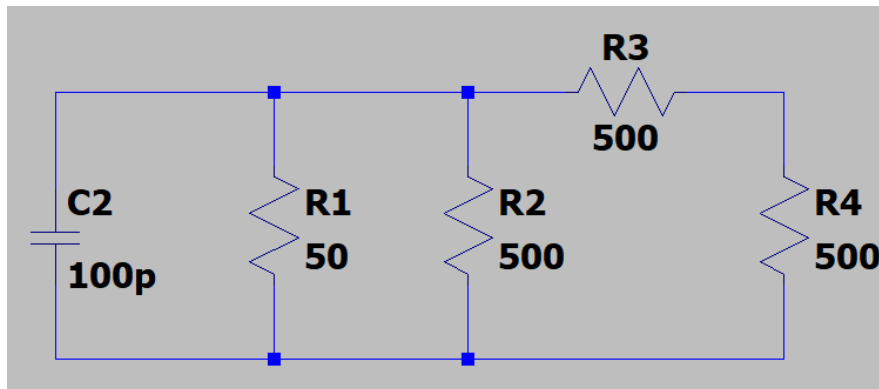


Circuit for C_3 , giving us $R_{th} = R1 // R2 + R3 + R4$

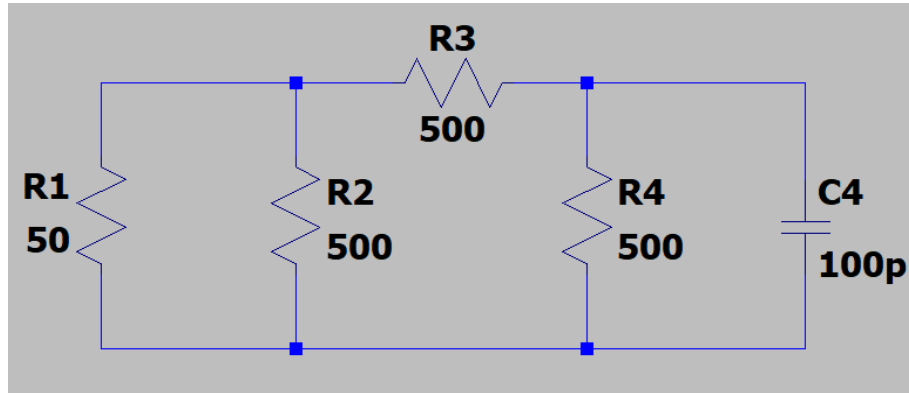
High frequency poles

Then we do the OC time constant test, shorting C_1 and C_3 , then getting the thevenin resistance for each capacitor whilst opening the other. The poles are once again $\omega_{p,Ci} = 1 / (R_{th}C_i)$, and this gives us $\omega_{p,C2} = 2.3E8$ and $\omega_{p,C4} = 2.83E7$. The resulting high frequency pole is

$\omega_H = \left(\frac{1}{\omega_{C2}^2} + \frac{1}{\omega_{C4}^2} \right)^{-\frac{1}{2}}$, which gives us $\omega_H = 3.781E7$ rad/s or 6.018 MHz.



Circuit for C_2 , giving us $R_{th} = R1 // R2 // (R3 + R4)$



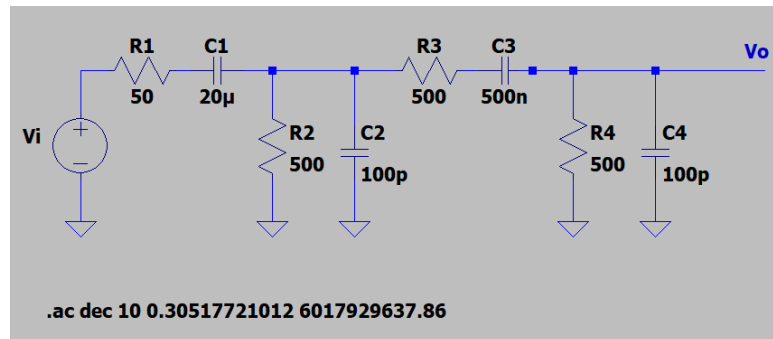
Circuit for C_4 , giving us $R_{th} = [(R1//R2) + R3] // R4$

wc1	130.434782609
wc3	1913.04347826
wl	1917.48496271
wc2	230000000.001
wc4	38333333.3332
wh	37811767.0802

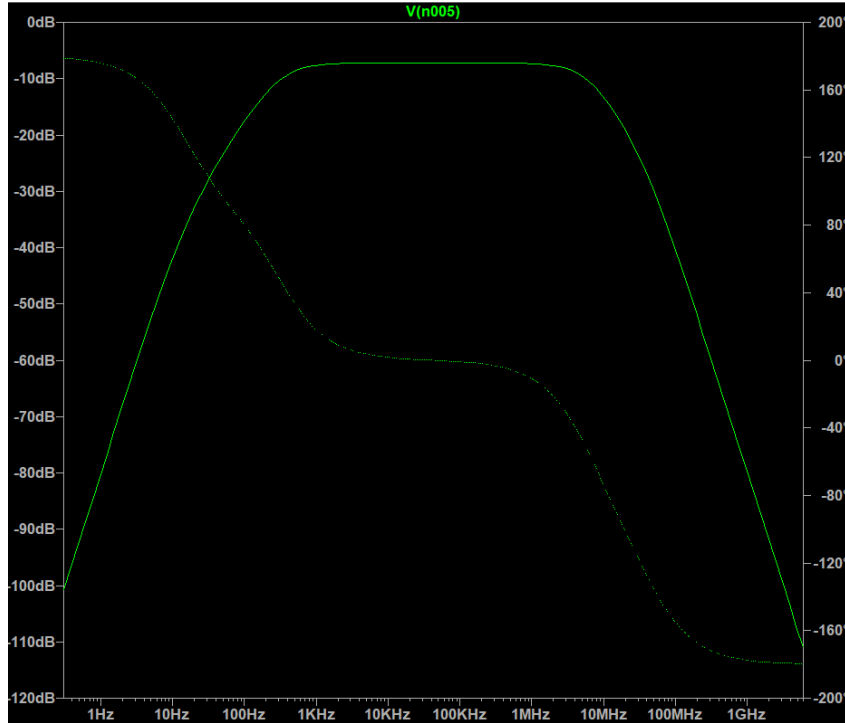
List of all the calculated values

LTspice Simulation

With our low and high frequency poles we can now set the appropriate bounds for our simulation on LTspice with three decades below ω_L (~ 0.3) and three decades above ω_H ($= 6E10$):



We get the following plot, with the solid line representing the magnitude and the dashed line showing the phase:



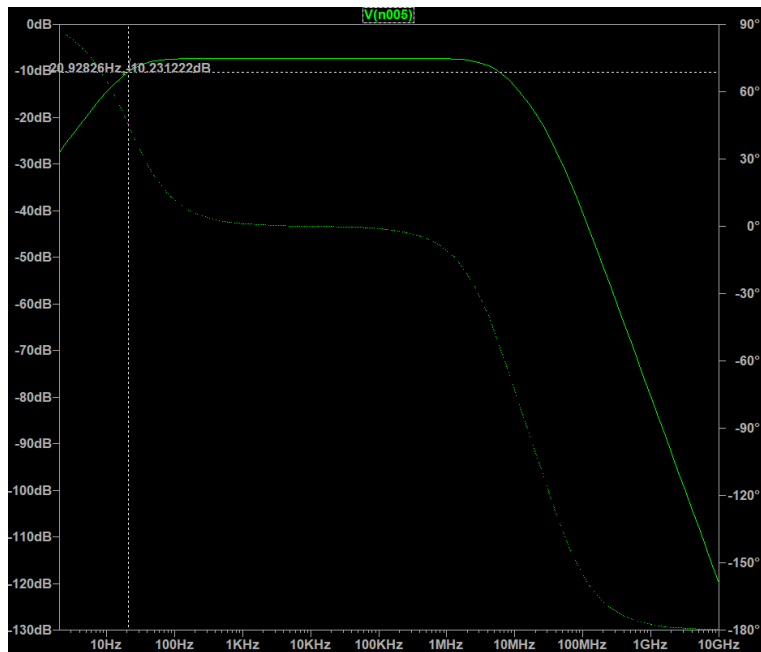
From this we observe the peak value is at approximately -7.2356dB, and we can graphically identify our high frequency and low frequency points as the points on the graph where the magnitude is roughly -10.2356dB, as marked below:



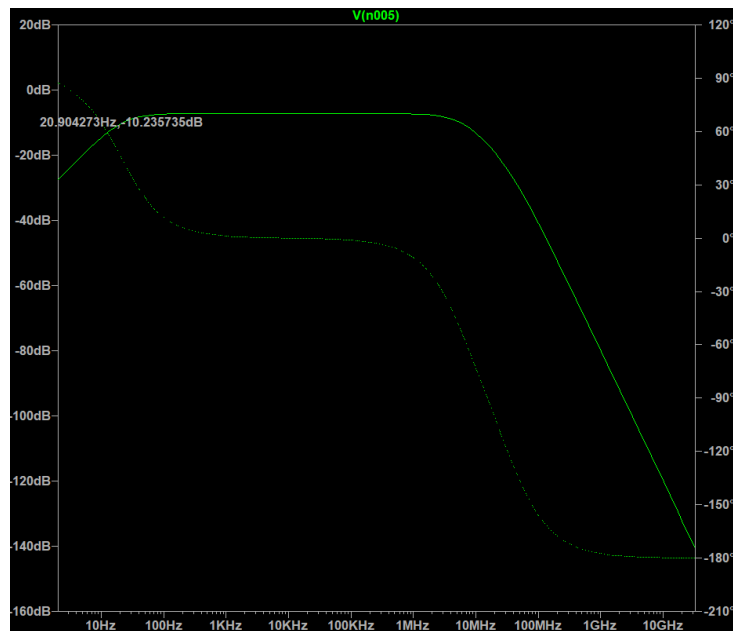
1B. Increase the value of C3 to 1mF, 2 mF, 5 mF, and 10 mF and rerun the AC simulation recording the effect on the low frequency poles for each C3 value. Determine the error percentage in the calculated low-frequency 3-dB point (calculated using the method of OC and SC time constants) as compared to those obtained from the simulation for each value of C3 value (including the initial C3 = 500 nF).

Rerunning Simulations

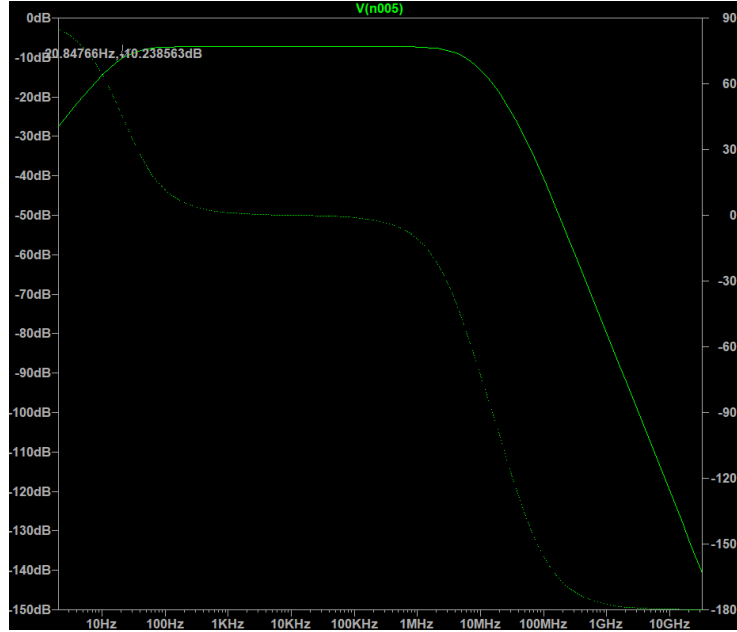
Rerunning the simulation for $C_3 = 1\text{mF}$:



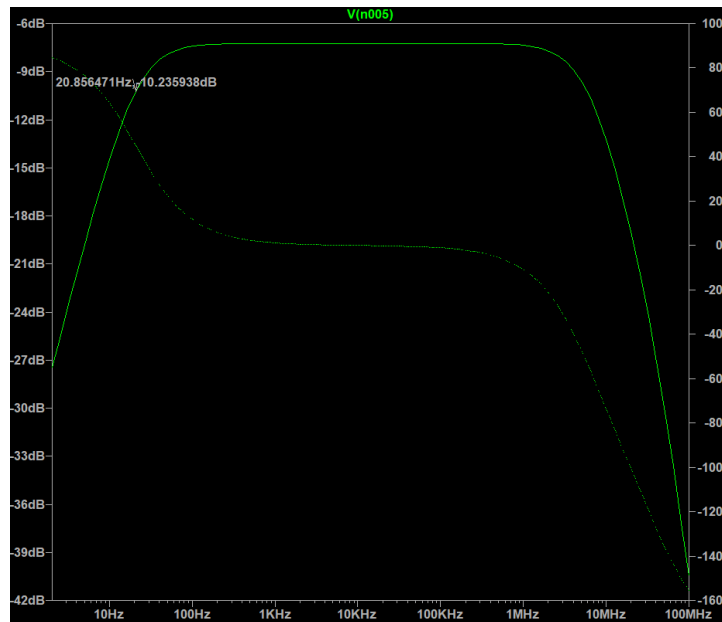
For $C_3 = 2\text{mF}$:



For $C_3 = 5\text{mF}$:



For $C_3 = 10\text{mF}$:



Error values

With all the plots and our graphically-found low frequency poles, we can now compute the error values between what we found on the graph and what we calculated with the OC and SC time constant methods. Since C_3 is the only element in the circuit that had changed, we can re-use the R_{th} value we found in part 1A) (which was 1045.45Ω) as well as the pole for C_1

(130.43 rad/s) to calculate the new pole in rad/s with the expression $\omega_L = \sqrt{\omega_{C1}^2 + \left(\frac{1}{R_{th,C3}C_3}\right)^2}$,

and then find that value in Hz with $\omega = 2\pi f$. We can then compare this value to what we found graphically, and calculated the accuracy with the formula $\% Error = \frac{\omega_{L,calculated} - \omega_{L,graphical}}{\omega_{L,graphical}} \cdot 100$

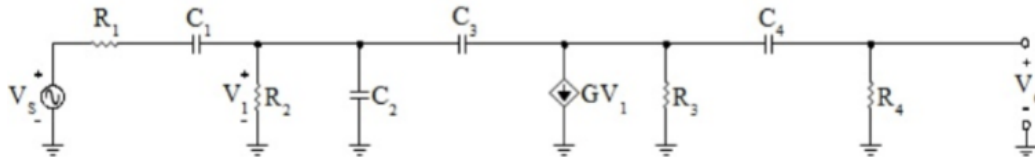
This gives us the following table of values:

C ₃ Value	500nF	1mF	2mF	5mF	10mF
Calculated ω_L (rad/s)	1917.48	131.39	130.9	130.63	130.53
Calculated ω_L (Hz)	305.17	20.91	20.86	20.79	20.77
Graphical ω_L	313.36	20.93	20.90	20.85	20.86
% Error	2.61%	-0.0956%	-0.191%	-0.288%	-0.431%

We see that the error is very small, and our approximations make for a good model of the system.

PROBLEM 2

P2. The figure below shows a basic transconductance amplifier. It has been designed with $R_1 = 50 \Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = R_4 = 2 \text{ k}\Omega$, $C_1 = 4 \mu\text{F}$, $C_2 = 10 \text{ pF}$, $C_3 = 2 \text{ pF}$, $C_4 = 2 \mu\text{F}$ and $G = 0.1 \text{ S}$.



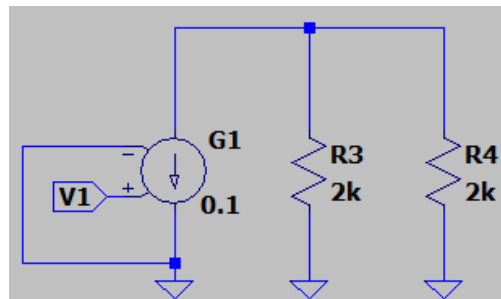
Basic transconductance amplifier

2A. Calculate the mid band gain and the locations of all of the poles (and any zeros) of the circuit P2 using Miller's theorem and the method of OC and SC time constants.

Applying Miller's Theorem

To analyze this circuit with the OC and SC time constants method, we must first apply Miller's Theorem to deal with the C_3 capacitor that filters high frequency.

To find the gain across the capacitor, we short the voltage source and low frequency capacitors (C_1 and C_4), and open our high frequency capacitors. This gives us the following circuit:

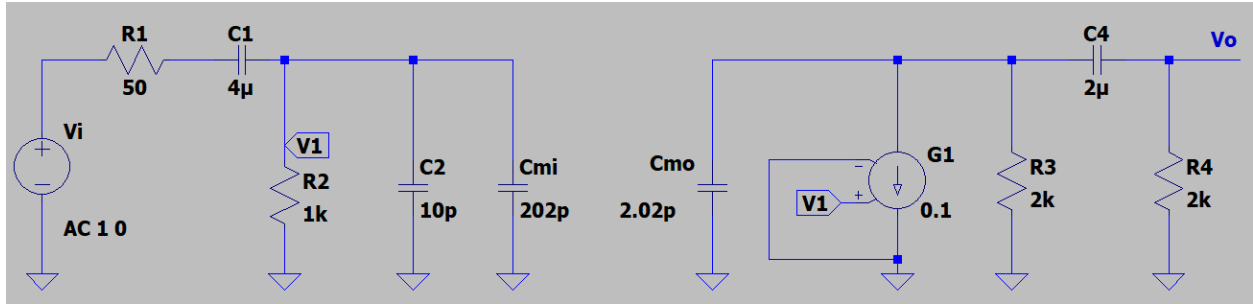


From this, we derive the equation: $0.1V_1 = \frac{0-V_2}{R_3} + \frac{0-V_2}{R_4}$

Solving for $\frac{V_2}{V_1}$ gives us the gain over C_3 , which results in $A_{C_3} = \frac{V_2}{V_1} = -100$.

Applying Miller's Theorem, we get the input capacitor $C_{mi} = \frac{Z_{C_3}}{A_{C_3} + 1} = C_3(A_{C_3} + 1) = 202 \text{ pF}$ and

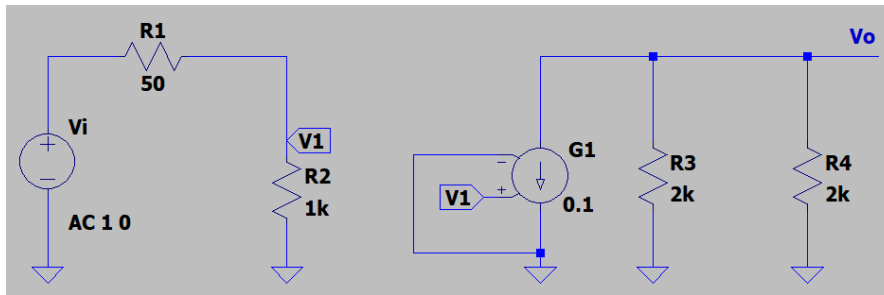
the output capacitor $C_{mo} = \frac{Z_{C_3} \cdot A_{C_3}}{A_{C_3} + 1} = \frac{C_3(A_{C_3} + 1)}{A_{C_3}} = 2.02 \text{ pF}$. The circuit then converts to:



Resulting circuit using Miller's Theorem

Midband Gain

First, to get the midband gain A_M , we simply short all the low frequency capacitors (C_1 , C_3 , and C_4) and open all the high frequency capacitors (C_2), then we find our equivalent transfer function for the circuit. Following this procedure we get the following circuit:



From here we derive the following equations

$$\frac{V_i - V_1}{R_1} = \frac{V_1}{R_2}, \text{ and } 0.1V_1 = \frac{0 - V_o}{R_3} + \frac{0 - V_o}{R_4}$$

From the latter we got our gain for applying Miller's Theorem which was

$$A_{C3} = \frac{V_o}{V_1} = -100, \text{ so solving the first of the two expressions for } \frac{V_o}{V_i} \text{ yields}$$

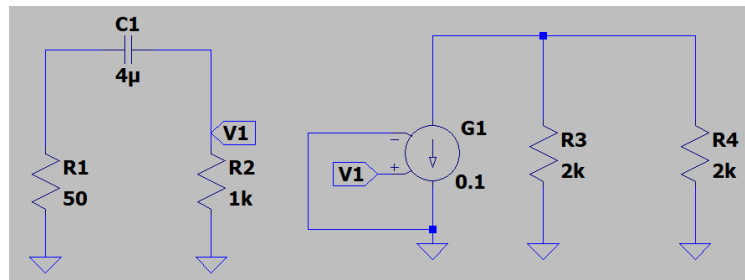
$$A_M = \frac{-2000}{21} = -95.238.$$

Low frequency poles

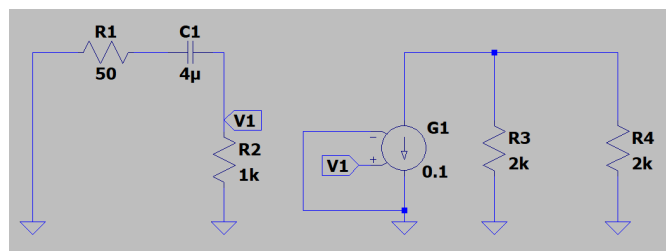
Next, we'll start by applying the SC time constant test, where we open our high frequency capacitors C_2 , C_{mi} and C_{mo} , and go find the thevenin resistance as well as respective poles for the remaining low frequency capacitors by shorting all but whichever one we are testing. Like in Problem 1, we use these 3 expressions:

- $\omega_{Ci} = \frac{1}{R_{th} C_i}$
- $\omega_L = \sqrt{\omega_{C1}^2 + \omega_{C3}^2}$
- $\omega = 2\pi f$

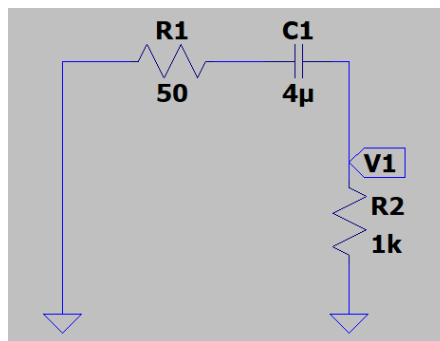
In this circuit, C_1 and C_4 are our low frequency capacitors. Starting with C_1 , we get this circuit:



To solve for the thevenin equivalent resistance, we can replace C_1 with a 1V DC voltage source, and then solve for the current going through said source. Doing so gives us the circuit below:



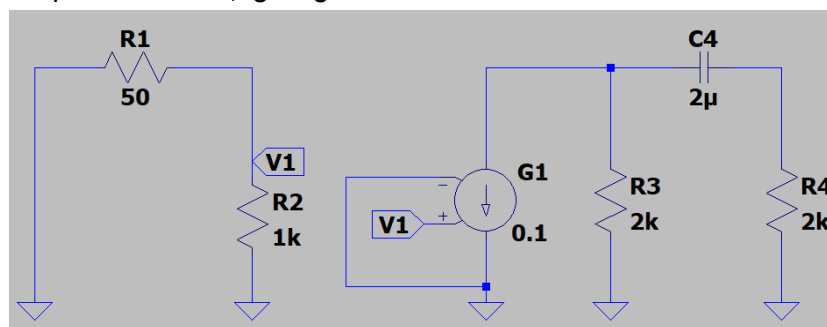
We can neglect the right half as it doesn't provide any useful information, so we end up with:



From this we see that $R_{th} = R_1 + R_2 = 1050 \Omega$, and so we find our pole to be

$$\omega_{C1} = \frac{1}{R_{th}C_1} = \frac{1}{(1050)(4 \cdot 10^{-6})} = 238.1 \text{ rad/s} = 37.894 \text{ Hz}.$$

Now we repeat the process for C_4 , giving us the circuit:



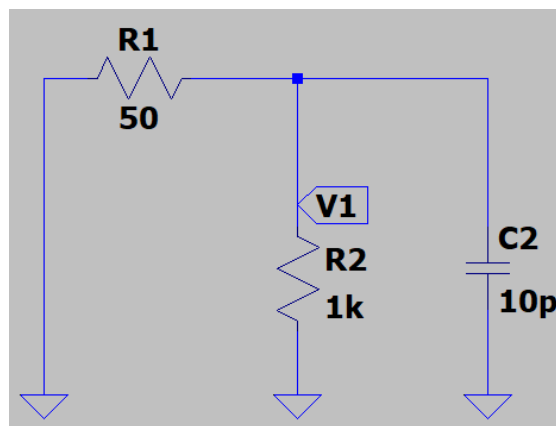
We see that $V_1 = 0V$, and so our current source is also 0 A. Thus from respect to C_4 we get $R_{th} = R_3 + R_4 = 4 k\Omega$, and so we find our pole to be

$$\omega_{C4} = \frac{1}{R_{th}C_4} = \frac{1}{(4000)(2 \cdot 10^{-6})} = 125 \text{ rad/s} = 19.894 \text{ Hz}.$$

Now we can find our low frequency pole $\omega_L = \sqrt{\omega_{C1}^2 + \omega_{C4}^2} = 268.91 \text{ rad/s} = 42.799 \text{ Hz}$

High frequency poles

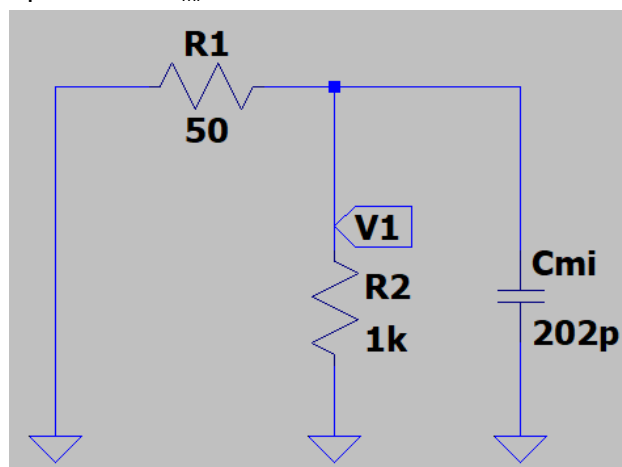
Moving on to the high frequency pole, we begin with C_2 , shorting the source and all low frequency capacitors, and opening the other high frequency capacitors. This gives us the following circuit:



We see $R_{th} = R_1 // R_2 = 47.62 \Omega$, and so we get

$$\omega_{C2} = \frac{1}{(47.62)(10 \cdot 10^{-12})} = 2.1 \cdot 10^9 \text{ rad/s} = 334.225 \text{ MHz}$$

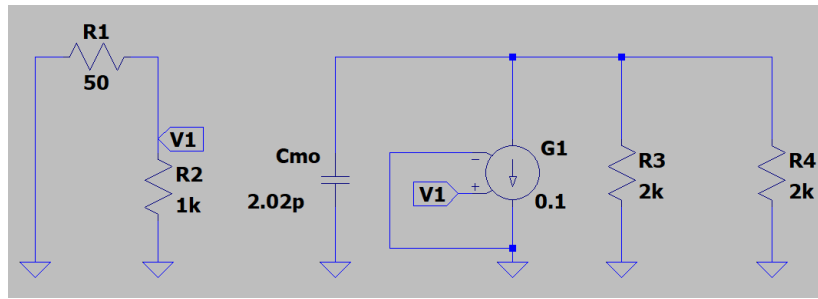
Next is our input Miller capacitance C_{mi} . With the same method we derive the circuit:



We find that $R_{th} = R_1 // R_2$ (same as for C_2), and we get the pole

$$\omega_{Cmi} = \frac{1}{(47.62)(202 \cdot 10^{-12})} = 1.0396 \cdot 10^8 \text{ rad/s} = 16.546 \text{ MHz}$$

Lastly is the output Miller impedance C_{mo} , which gives us the circuit:



V_1 is 0V, and so our controlled current source becomes an open circuit. We see that with respect to C_{mo} , $R_{th} = R_3 \parallel R_4 = 1 \text{ k}\Omega$, and thus we get

$$\omega_{Cmo} = \frac{1}{(1000)(2.02 \cdot 10^{-12})} = 4.9505 \cdot 10^8 \text{ rad/s} = 78.790 \text{ MHz}$$

ω_{Cmi} is the smallest of our poles and so is the dominant high frequency pole. Still, for accuracy's sake we can calculate the full high frequency pole to be

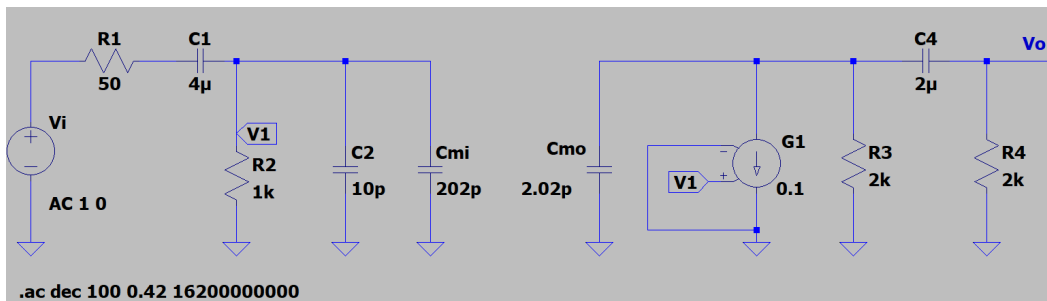
$$\omega_H = \left(\frac{1}{\omega_{C2}^2} + \frac{1}{\omega_{Cmi}^2} + \frac{1}{\omega_{Cmo}^2} \right)^{-\frac{1}{2}} = 1.016 \cdot 10^8 \text{ rad/s} = 16.174 \text{ MHz}$$

2B. Run an AC simulation on the circuit over a frequency range from at least 3 decades below the low frequency 3-dB point to at least 3-decades above the high frequency 3-dB point. Plot the Bode plots, both magnitude and phase, for the circuit. Identify and record the location of each pole (and any zeros) of the transfer function using the magnitude plot and compare these with the values calculated in part A, above. Also, calculate the error percentage in the estimated 3-dB points (calculated using the values of the pole locations obtained in part A, above) as compared to those obtained from the simulation.

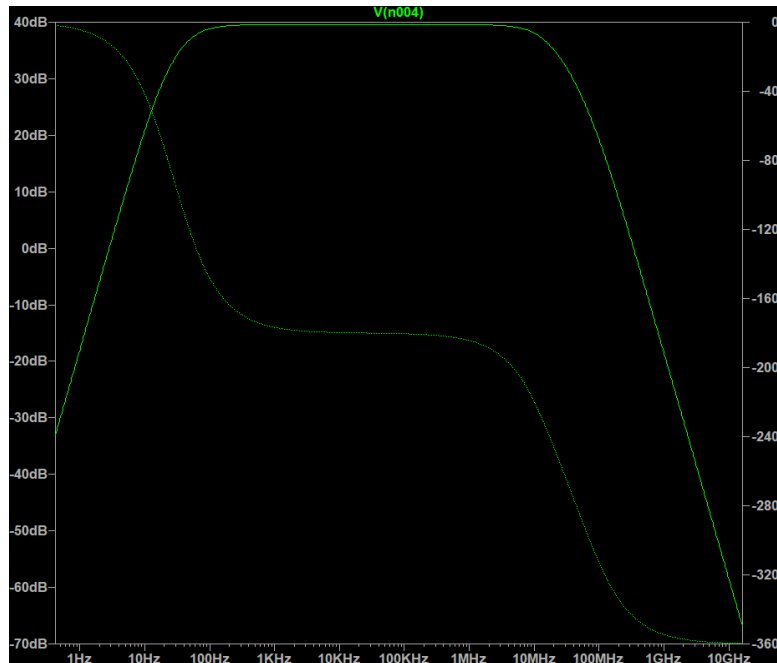
Simulating the circuit

With our frequency cutoffs found, we can run the circuit from just over 3 decades below our low frequency cutoff, or 0.042 Hz, and 3 decades above our high frequency cutoff, at 16.2 GHz.

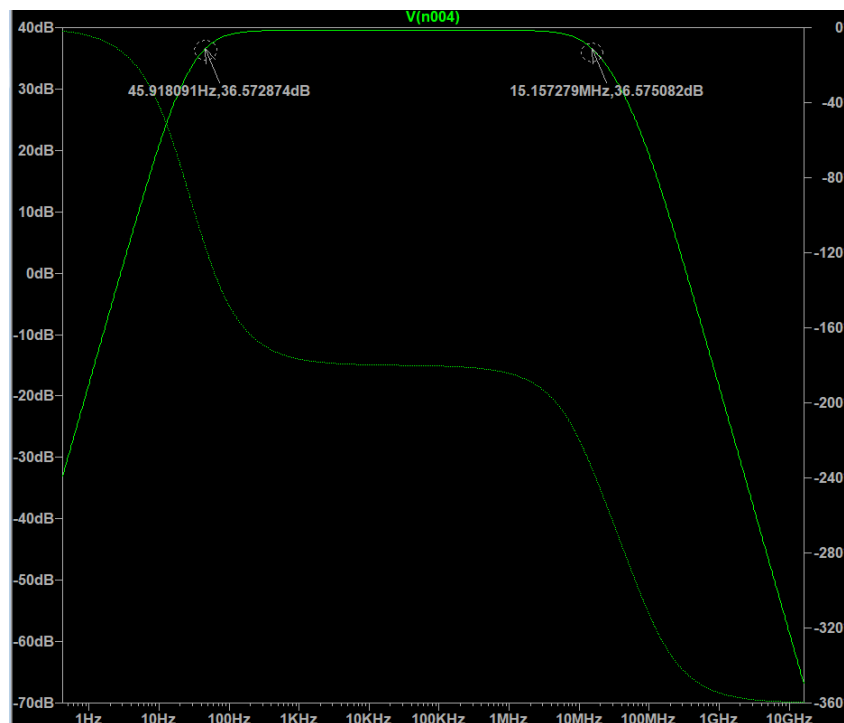
Using LTspice, we model the circuit as such:



Running our simulation, we get the following Bode plots:



We find our peak magnitude to be at 39.576 dB, and so our cutoff frequencies will be the points 3 dB below that, marked below:



Low frequency cutoff @ 45.92 Hz, High frequency cutoff @ 15.16 MHz

Error calculation

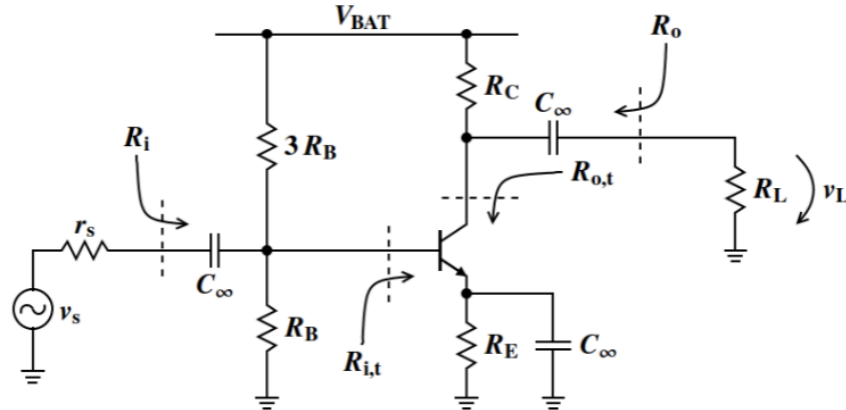
Using the formula $\% \text{ Error} = \frac{\omega_{\text{calculated}} - \omega_{\text{graphical}}}{\omega_{\text{graphical}}} \cdot 100$, we derive the following table of values:

Pole	Calculated value (Hz)	Graphical value (Hz)	% Error
ω_L	42.80	45.92	-6.79%
ω_H	16174000	15160000	6.69%

We see our calculated values are not too far off from what we found on the graph.

PROBLEM 3

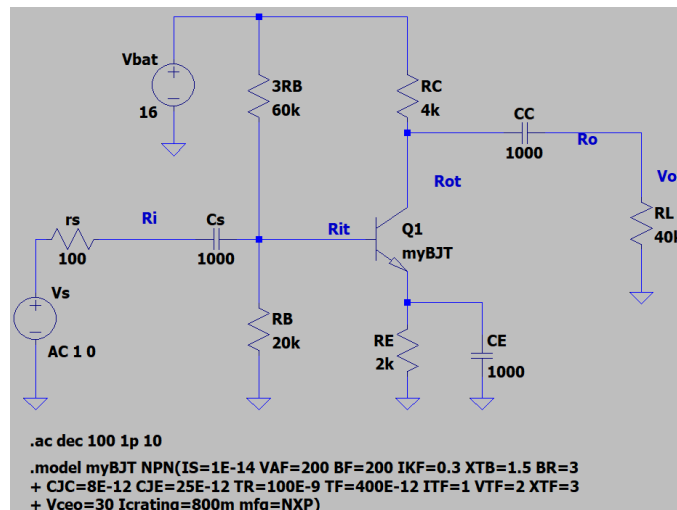
P3. We consider the following circuit, with the following numerical values: $V_{BAT}=16V$ (supply voltage), $R_B=20k\Omega$, $R_C=4k\Omega$, $R_E=2k\Omega$, $r_s=100\Omega$, $R_L=40k\Omega$ (load resistance), $C_\infty \rightarrow \infty$ (coupling/bypass capacitors), $\beta=200$, $V_A=200V$ (Early voltage).



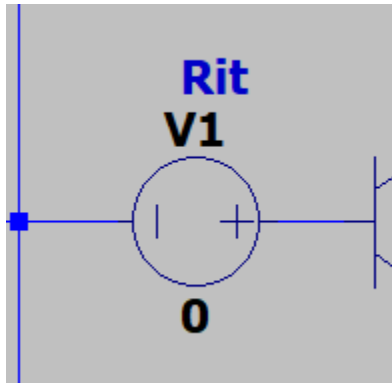
3A. Introduce the circuit in Spice and simulate and separately calculate (in midband) the input resistances R_i , R_{it} and the output resistances R_o , R_{ot}

Spice simulation

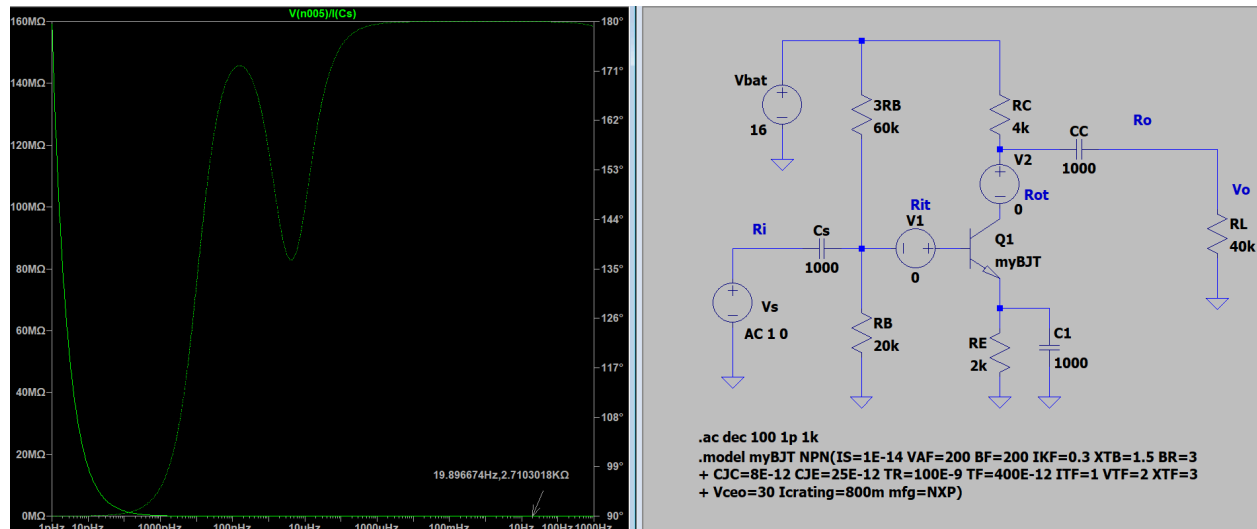
To model this circuit in LTspice, we can replace the infinitely large capacitors with an incredibly large value. We must be cautious not to put too large of a value otherwise the program is unable to properly simulate the circuit and will give inaccurate results, such as by treating the capacitor as a short since it is unable to properly compute its value. For this I settled on a capacitance of 1 kF for the infinitely large capacitors and ran the simulation between the frequencies of 1 pF and 10 F. We also need to decide on the parameters for our BJT. For this I copied the preset conditions for a 2N222 transistor, removed the internal resistances, and set the Early voltage and Beta value to both be 200, resulting in the following:



To get our desired resistances, we essentially neglect all components away from wherever the arrow is facing, apply a voltage source at the input or output point, and then divide the voltage source by the current going through it at a point along the midband to figure out the equivalent resistance. LTspice makes this calculation easy as we can directly divide the values onto the plot. Since R_{it} and R_{ot} don't have any components at their nodes, we can introduce a 0V source that will let us read the current across it as such:

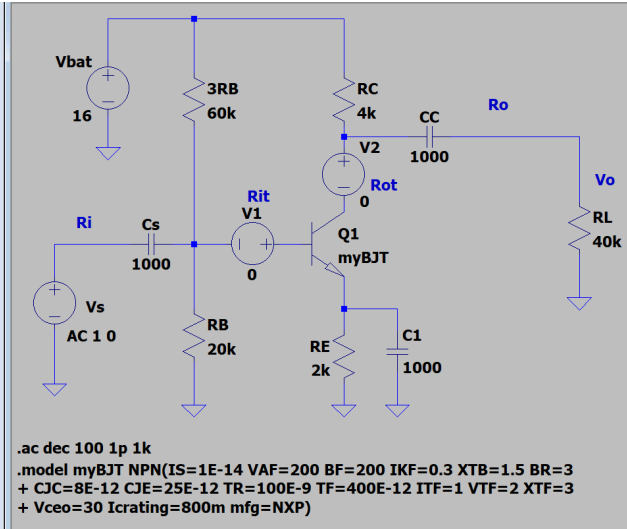
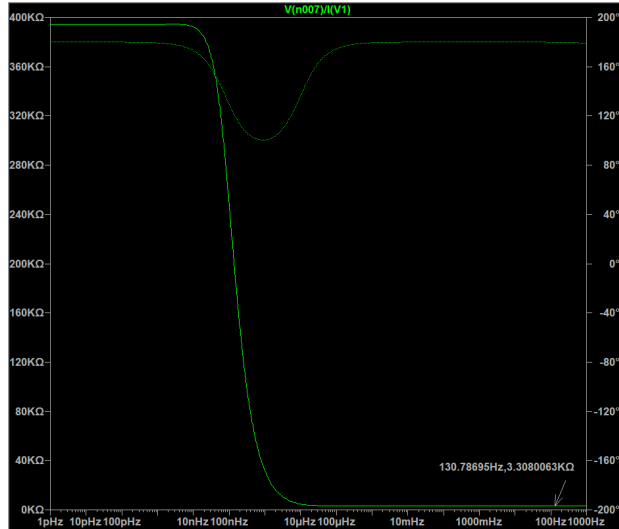


Starting with R_i , we get the following circuit and plot:



Plotting the resistance we get $R_i = 2710.30 \Omega$.

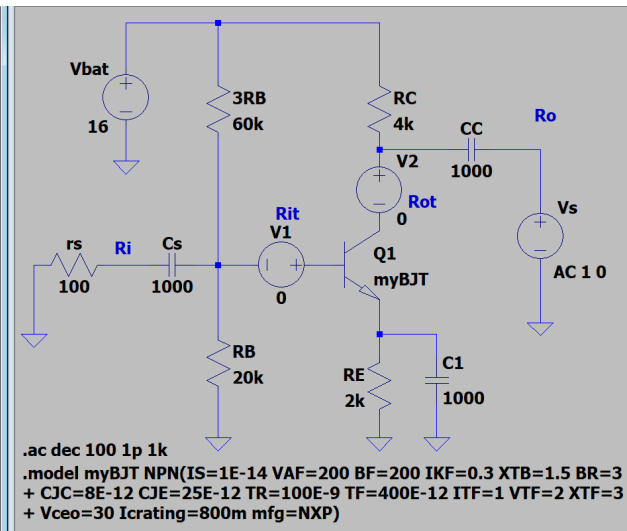
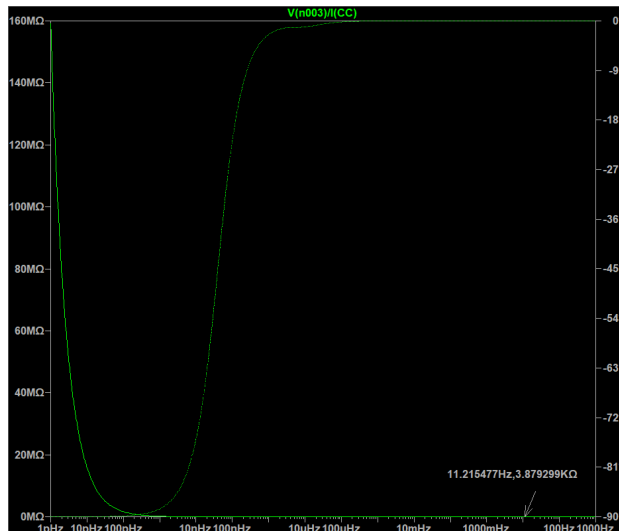
For R_{it} :



We find that $R_{it} = 3308.01 \Omega$.

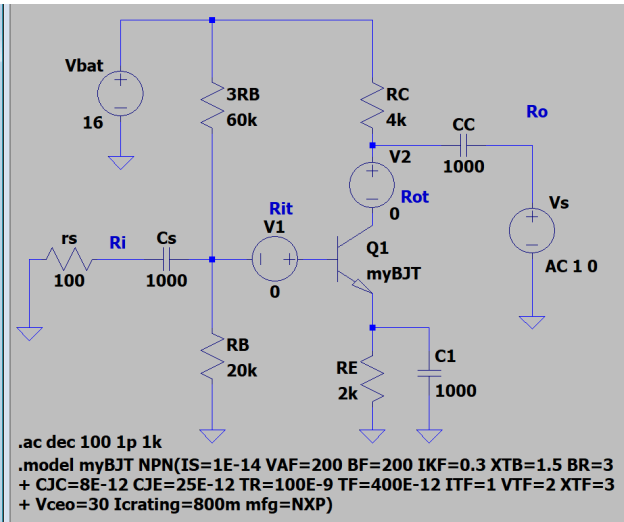
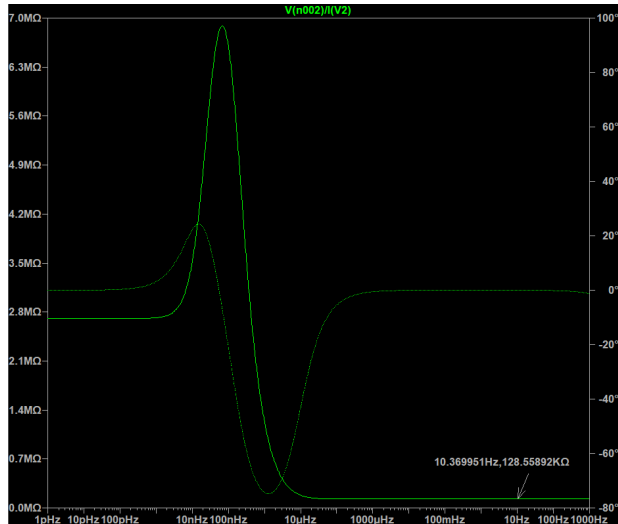
For the output resistances, we reset the circuit and move the voltage source V_s to the R_o node, while removing the R_L resistor as we don't include it in our calculation.

For R_o , plotting the voltage over the current through the large capacitor we get:



This gives us $R_o = 3879.30 \Omega$

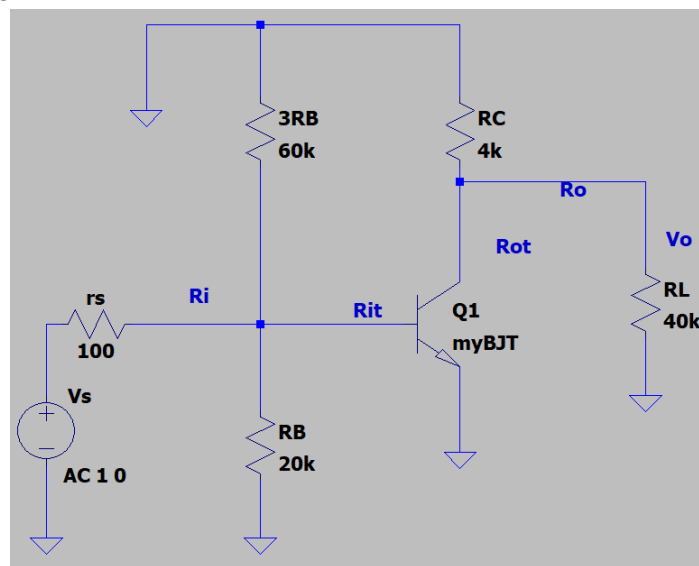
Lastly for $R_{o, it}$, with our 0V source to help us plot the current we get:



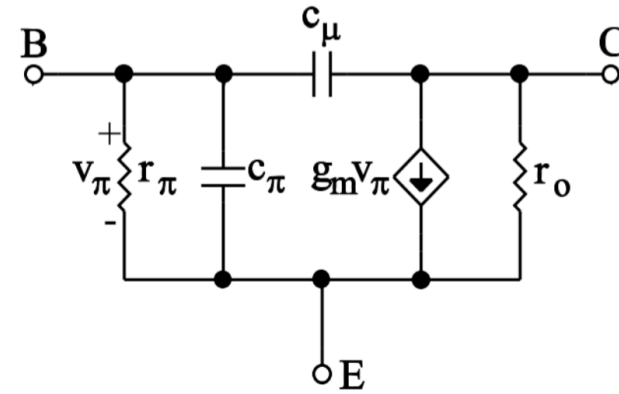
This gives us $R_{ot} = 128.559 k\Omega$

Separate calculations of resistances R_i , R_{it} , R_o , and R_{ot}

Since we're operating in the midband, we can short our low frequency capacitors. This gives us the resulting circuit:



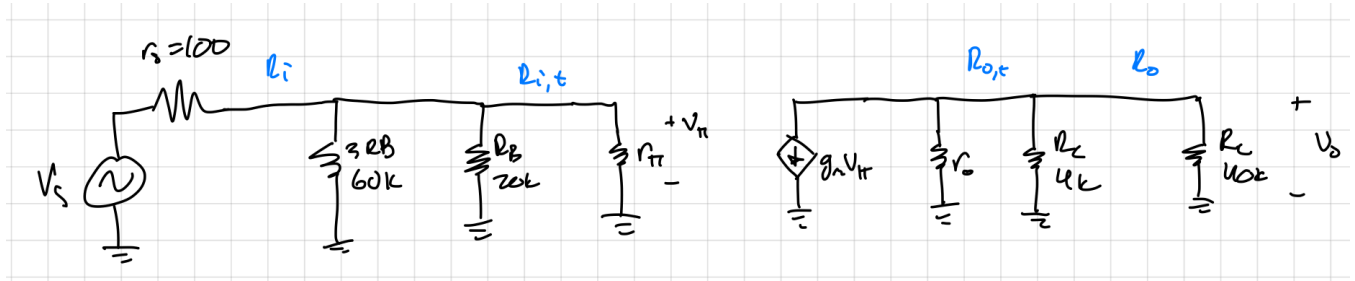
From here we can model our BJT with the AC small signal model



Source: ELEC 301 slides

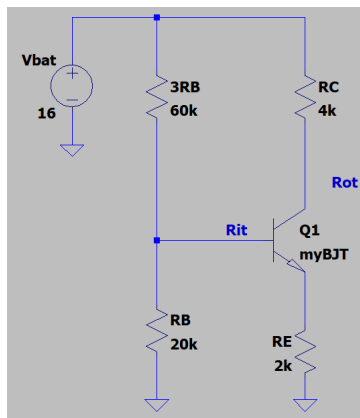
Where $r_{\pi} = \frac{V_T}{I_B} = \beta \frac{V_T}{I_C}$, $r_o = \frac{V_A}{I_C}$, and $g_m = \frac{I_C}{V_T}$. We assume $V_T = 25\text{mV}$ @ 25°C .

Since we're in the midband, we open c_{π} , then apply Miller's theorem on c_{μ} before opening the resulting capacitors connected to ground. Substituting this into the circuit gives:

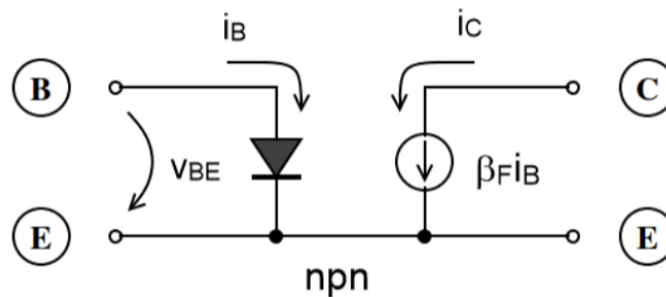


Before proceeding we need to calculate the values of our BJT small signal model components. To do this, we need to know the DC operating conditions of the circuit.

In DC, the low frequency resistors block any DC signal and so we get:

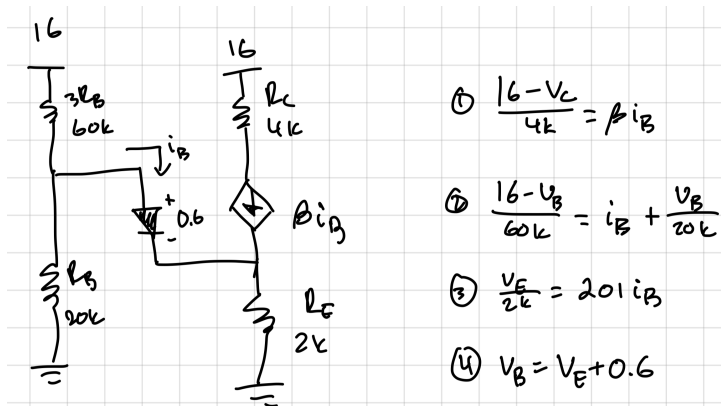


Simplified circuit



Large signal model from ELEC 301 slides

Combining the two and assuming we are operating in the active mode, we get the following circuit along with the listed equations:



With 4 equations and 4 unknowns, we're able to solve with a calculator and get:

vb	3.87769784173
vc	9.47721822542
ve	3.27769784173
ib	8.15347721823E-6

Our current I_B is positive and $V_C - V_B > 0$, therefore we were correct in our assumption that we are in the active mode and can be confident with the values we found. We can also derive that $I_C = \beta I_B = 1.631 \text{ mA}$

Returning to our small signal model, we can now calculate the values of our components:

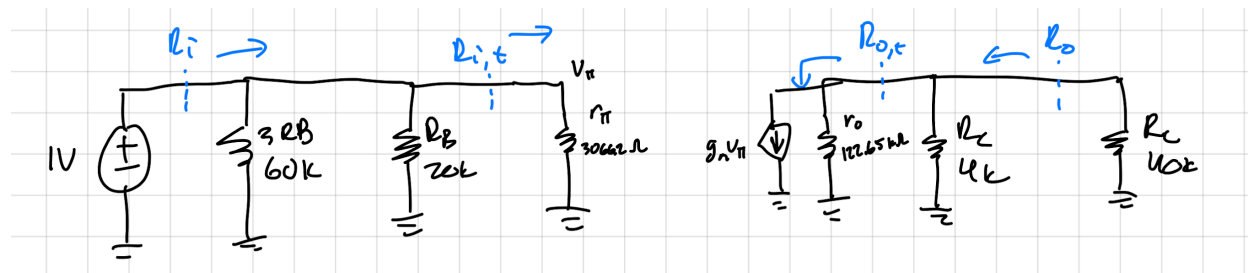
$$r_{\pi} = \frac{V_T}{I_B} = \frac{0.025}{8.15347722E-6} = 3066.176 \Omega$$

$$r_o = \frac{V_A}{I_C} = \frac{200}{1.63069544E-3} = 122.647 \text{ k}\Omega$$

$$g_m = \frac{I_C}{V_T} = \frac{1.63069544E-3}{0.025} = 0.065228$$

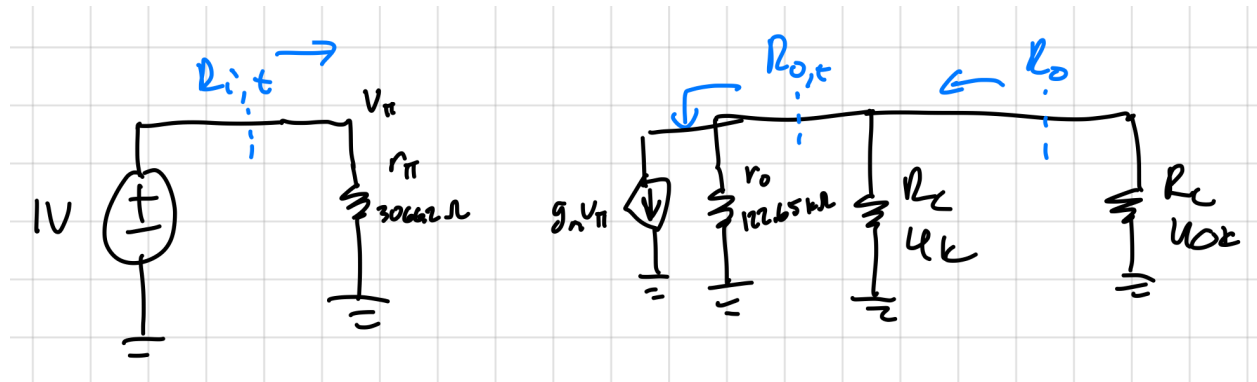
From here, we look at each of our desired resistances and, neglecting any components not included in the impedance, apply a voltage source and determine the current (if necessary) to figure out the equivalent resistance.

Starting with R_i , we remove r_s as it isn't affecting the resistance and apply a voltage source:



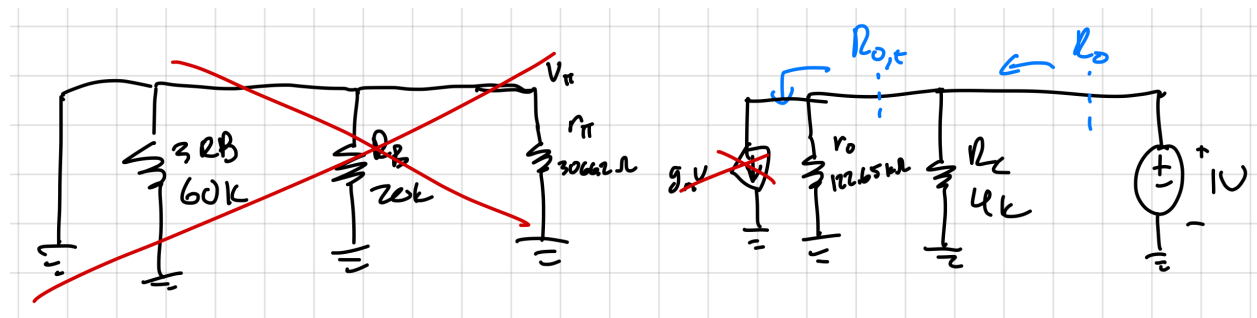
We see that R_i is unaffected by the right side, and so we simply get:
 $R_i = 3R_B // R_B // R_{\pi} = 2545.8 \Omega$

For $R_{i,t}$



This gives us $R_{i,t} = R_{\pi} = 3066.2 \Omega$

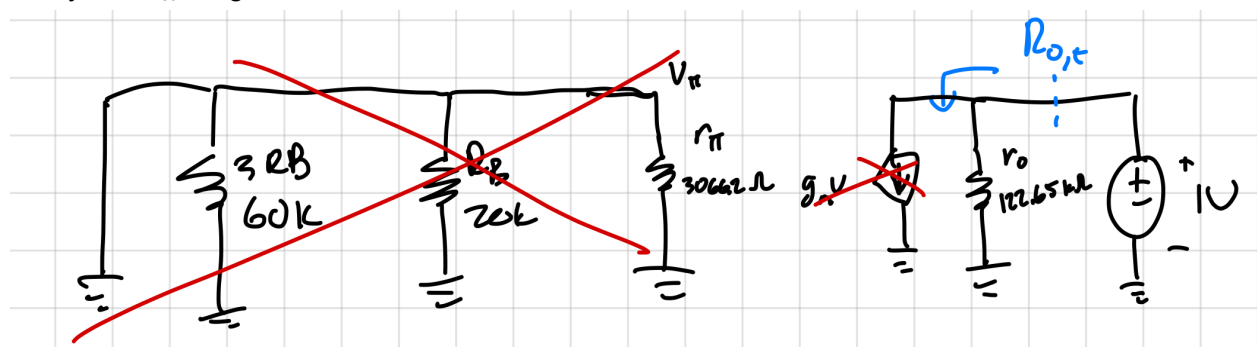
For R_o



Shorting V_s results in shorting the entire left side, and this $v_{\pi} = 0$. Therefore, we end up with

$$R_o = r_o // R_C = 3873.66 \Omega$$

Lastly, for R_{ot} we get



$$R_{ot} = r_o = 122.85 \text{ k}\Omega$$

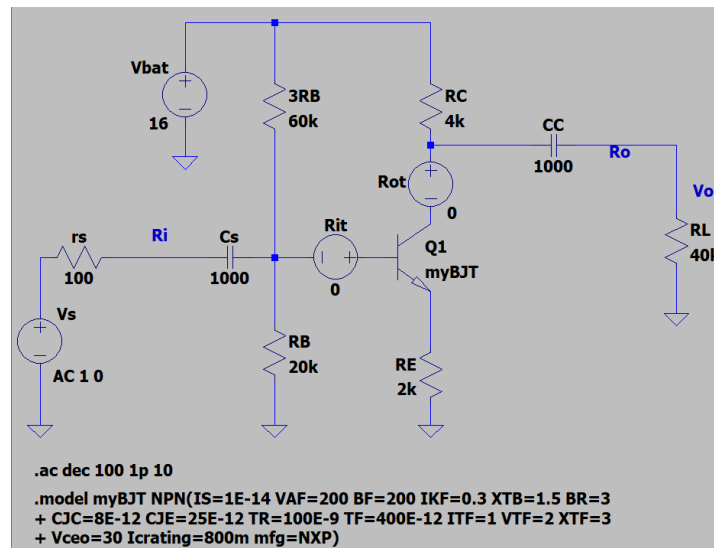
Comparing our values, we see that our calculations are quite close to the simulations:

Resistor	Simulated Value (Ω)	Calculated Value (Ω)	Error
R_i	2710.30	2545.8	-6.07%
$R_{i,t}$	3308.01	3066.2	-7.31%
$R_{o,t}$	3879.30	3873.66	-0.14%
R_o	128559	122850	-4.44%

3B. Remove $C_E = C_\infty$ from the circuit (so that R_E is not shunted anymore for the ac analysis). Simulate and recompute (in midband) the input resistances R_i , $R_{i,t}$ and the output resistances R_o , $R_{o,t}$. Comment upon how the presence of R_E affects the input and output resistances.

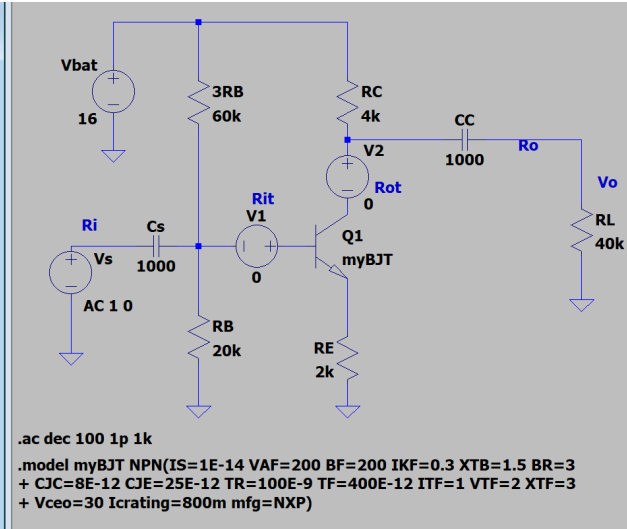
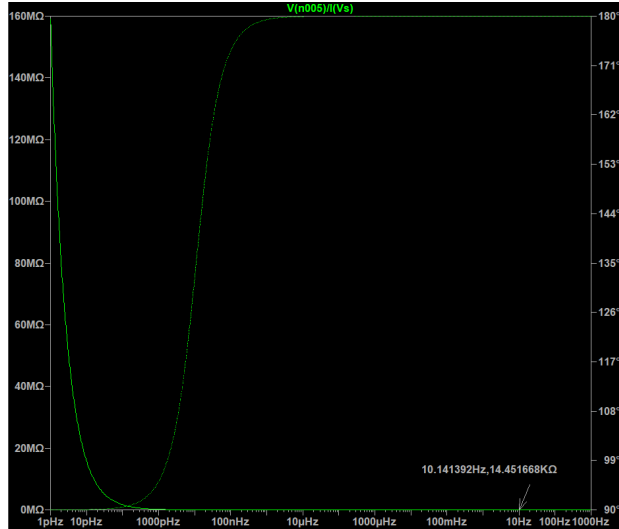
Spice simulation

Removing C_E and returning the 0V sources to simplify our measurements gives us:

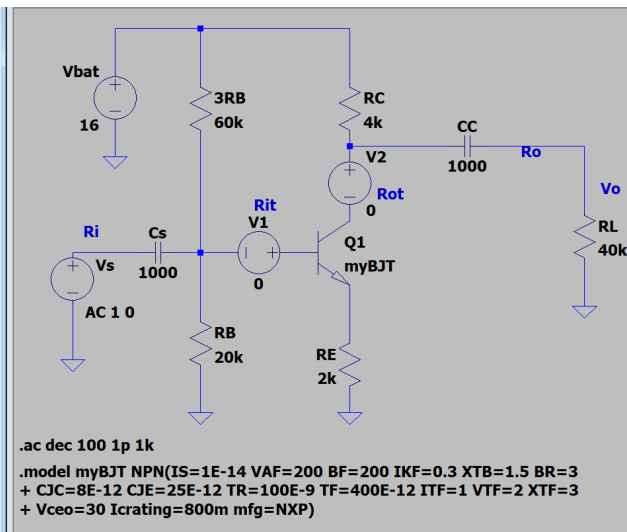


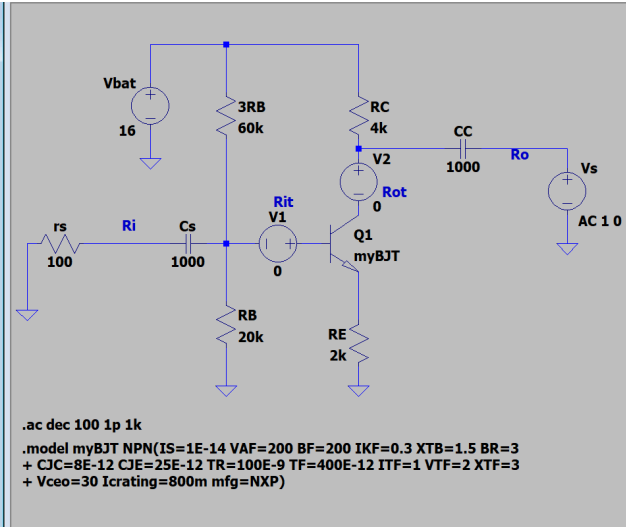
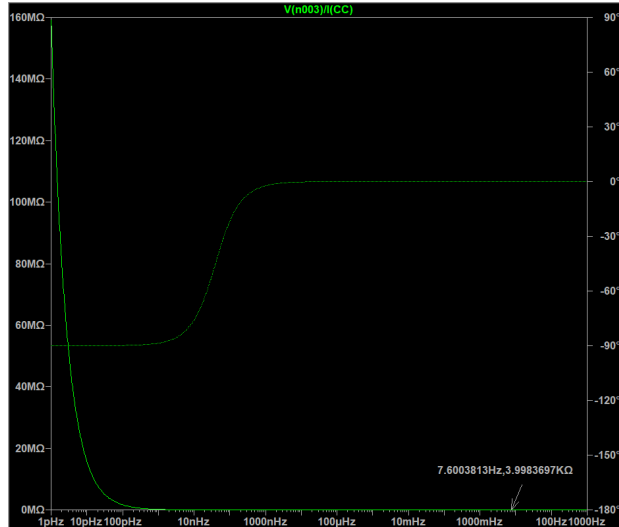
We repeat the same process as last time:

Starting with R_i , we get the following circuit and plot:



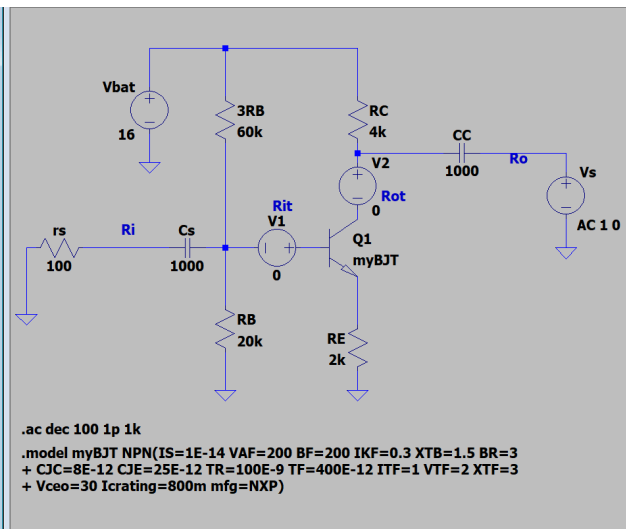
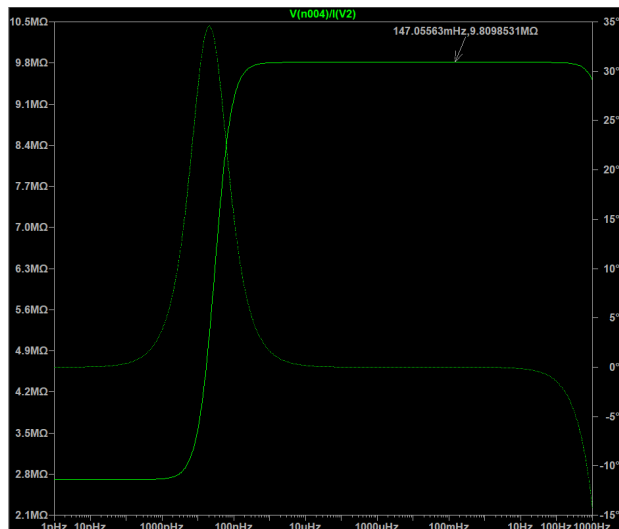
For R_{it} :





This gives us $R_o = 3998.37 \Omega$

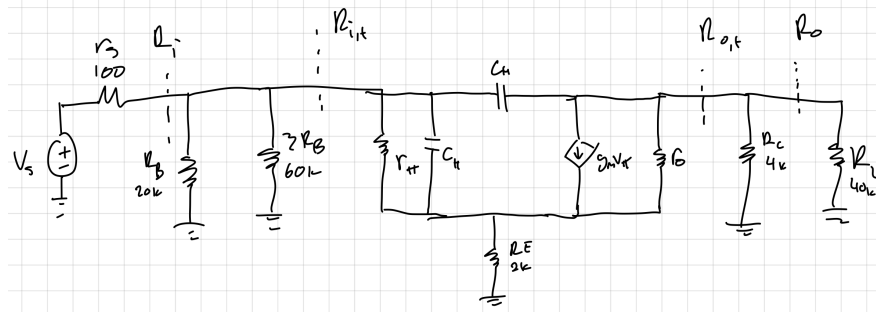
Lastly for $R_{o,t}$, we get:



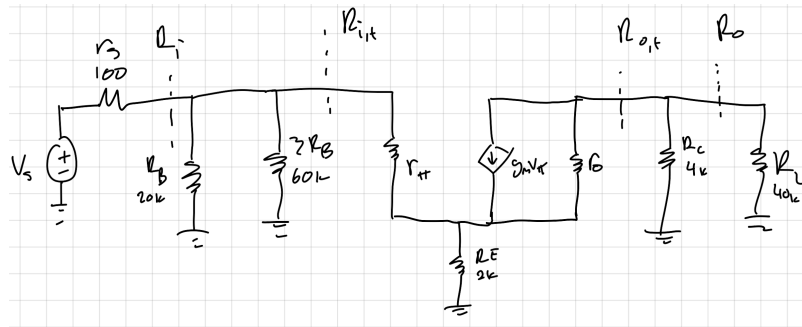
This gives us $R_{ot} = 9.80985 M\Omega$

Hand Calculations

First, we can redraw the small signal circuit, accounting for R_E no longer being shunted:



We see that the application of Miller's theorem on the c_μ no longer simply connects to ground and instead results in this circuit:



Removing the capacitor had no effect on the DC circuit, so we can reuse the previous values found in 3A:

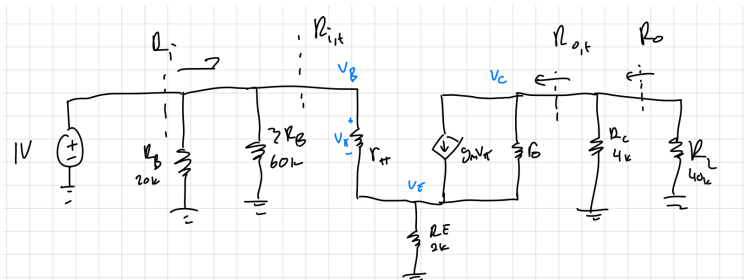
vb	3.87769784173
vc	9.47721822542
ve	3.27769784173
ib	8.15347721823E-6

This means that our r_π and r_o values stay the same and so:

$$r_\pi = 3066.176\Omega, r_o = 122.647\text{ k}\Omega, \text{ and } g_m = 0.065228$$

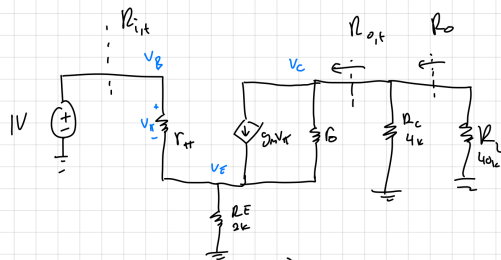
Just as in part 3A, we place a 1V source at the node the resistance starts at, neglect components facing away from the desired direction, and use KCL to solve for 1V over the current at the voltage source.

For R_{ia} :



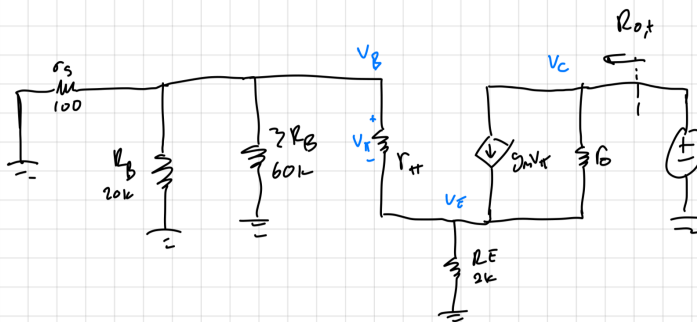
$$\begin{aligned} \textcircled{1} \quad I_i &= \frac{1}{R_B // R_E} + \frac{1 \cdot v_b}{r_\pi} \\ \textcircled{2} \quad v_\pi &= 1 - v_b \\ \textcircled{3} \quad \frac{1 - v_b}{r_\pi} + g_m v_\pi + \frac{v_c - v_b}{r_o} &= \frac{v_b}{R_E} \\ \textcircled{4} \quad \frac{0 - v_c}{R_C // R_L} &= g_m v_\pi + \frac{v_c - v_b}{r_o} \end{aligned} \quad \left. \begin{array}{l} R_i = \frac{1}{I_i} \\ \text{ } \end{array} \right\} R_i = 14.44\text{ k}\Omega$$

For R_{it} :



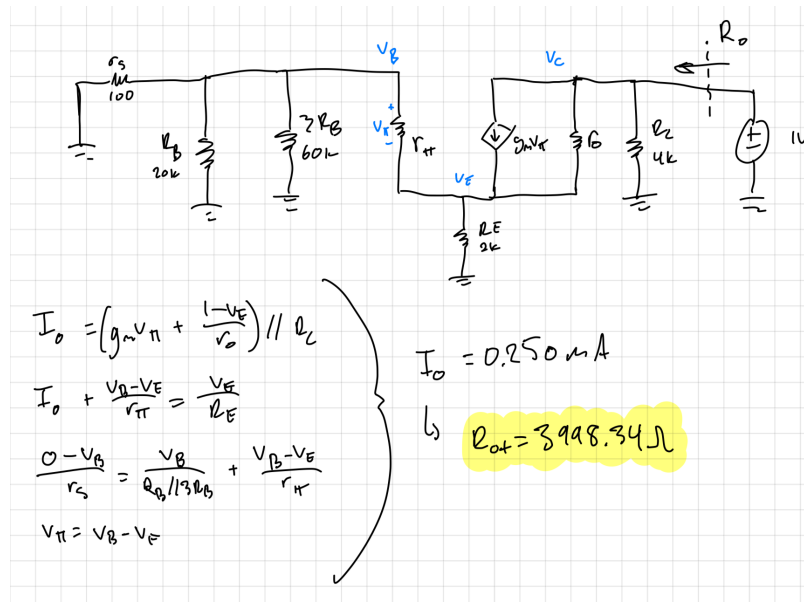
$$\begin{aligned}
 (1) \quad I_i &= \frac{1 - V_E}{r_{\pi}} \\
 (2) \quad V_E &= 1 - V_E \\
 (3) \quad \frac{1 - V_E}{r_{\pi}} + g_m V_E + \frac{V_C - V_E}{r_o} &= \frac{V_E}{R_E} \\
 (4) \quad \frac{0 - V_C}{R_C || R_L} &= g_m V_E + \frac{V_C - V_E}{r_o}
 \end{aligned}
 \left. \vphantom{\begin{aligned} (1) \\ (2) \\ (3) \\ (4) \end{aligned}} \right\} R_{i_k} = \frac{1}{I_{i_k}} = 387.460 \text{ k}\Omega$$

For R_{ot} :



$$\begin{aligned}
 I_{ot} &= g_m V_E + \frac{1 - V_E}{r_o} \\
 I_{ot} + \frac{V_E - V_E}{r_{\pi}} &= \frac{V_E}{R_E} \\
 \frac{0 - V_B}{r_s} &= \frac{V_B}{R_B || 3R_B} + \frac{V_B - V_E}{r_{\pi}} \\
 V_E &= V_B - V_E
 \end{aligned}
 \left. \vphantom{\begin{aligned} I_{ot} \\ I_{ot} \\ \frac{0 - V_B} \\ V_E \end{aligned}} \right\} I_{ot} = 0.104 \mu\text{A} \\
 \hookrightarrow R_{ot} &= 9.62125 \text{ M}\Omega$$

And lastly for R_o :



Comparison to 3A

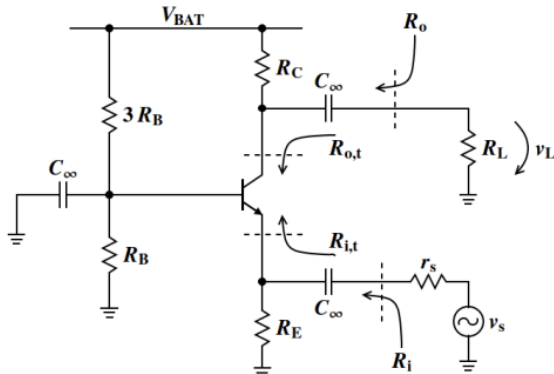
Comparing the simulated values of the two cases we get the following table:

Resistor	Simulated Value With Capacitor (Ω)	Simulated Value Without Capacitor (Ω)	Percent Difference(%)
R_i	2710.30	14452	533%
$R_{i,t}$	3308.01	395335	11950%
$R_{o,t}$	3879.30	3998.37	103%
R_o	128559	9809850	7631%

We see that removing the capacitor drastically increased all the resistances except for $R_{o,t}$ which only increased by 3%. R_i and R_o saw especially large differences. Overall shunting the R_E resistor substantially reduces the load.

PROBLEM 4

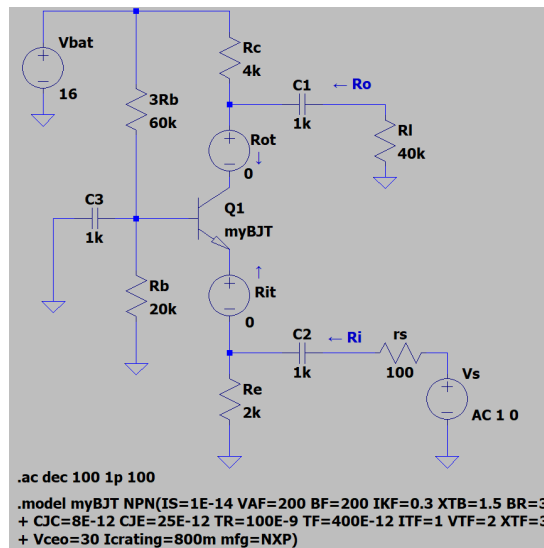
P4. We consider the common-base amplifier stage below, with the following numerical values: $V_{BAT}=16V$ (supply voltage), $R_B=20k\Omega$, $R_C=4k\Omega$, $R_E=2k\Omega$, $r_s=100\Omega$, $R_L=40k\Omega$ (load resistance), $C_{\infty} \rightarrow \infty$ (coupling/bypass capacitors), $\beta=200$, $V_A=200V$ (Early voltage). v_s is a small-signal voltage source.



4A. Introduce the circuit in Spice and simulate and separately calculate (in midband) the input resistances R_i , R_{it} and the output resistances R_o , R_{ot} . Make a comparison with the results obtained for the (3A) case.

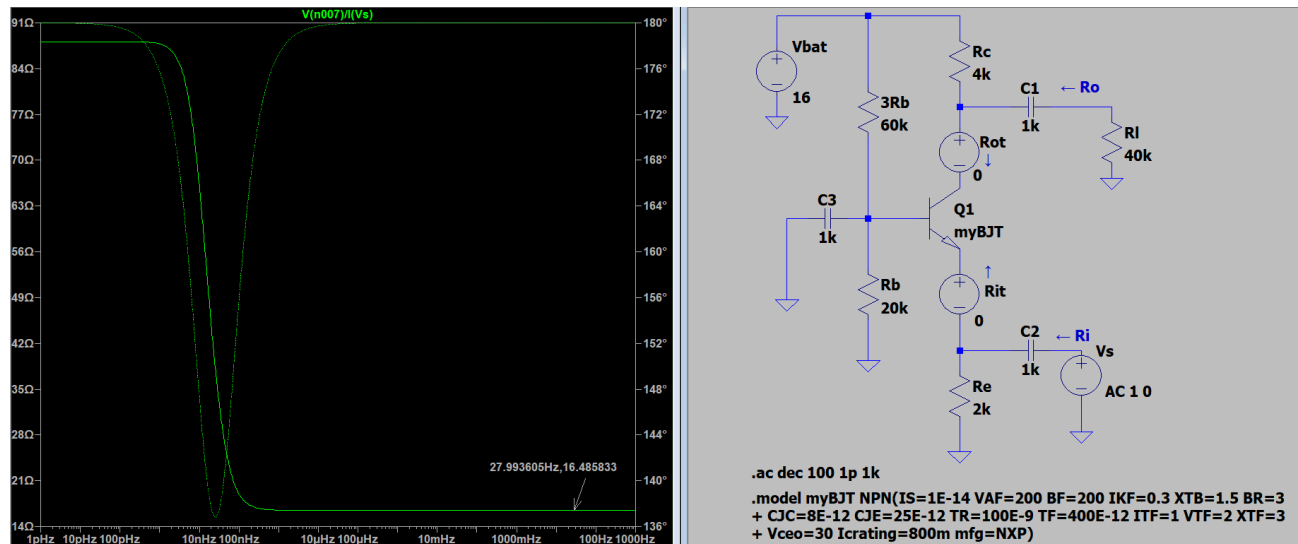
Simulation in LTspice

First we repeat the process of replacing the C_{∞} in the circuit with 1 kF capacitors (effectively ∞ with respect to the currents we're working with) and we'll reuse the BJT model defined in part 3 to get our exact copy of the circuit:



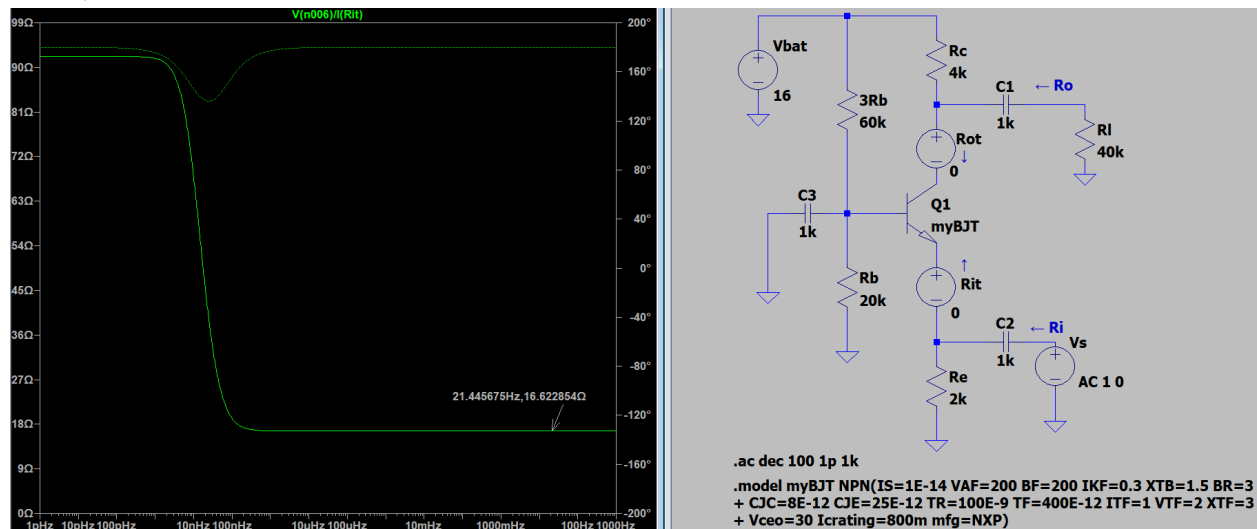
From here, we repeat the steps from part 3 of getting each resistance by removing any components that don't contribute to the desired direction, applying a 1 V source at the node, and finally finding the resistance by dividing said voltage by the current across the voltage source.

For R_i we get the following circuit and plot combination:



And find that $R_i = 16.486 \Omega$

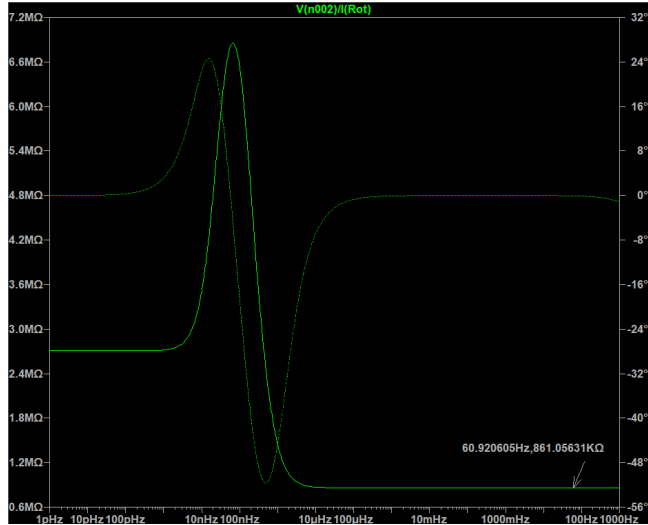
For $R_{i,t}$ keep the same circuit to get the proper DC values and get:



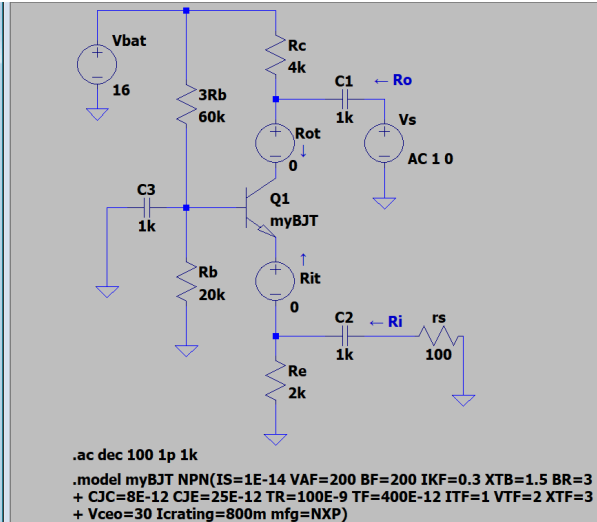
Giving $R_{i,t} = 16.623 \Omega$

For the output resistances, we move the voltage source to the R_o node instead and neglect R_L .

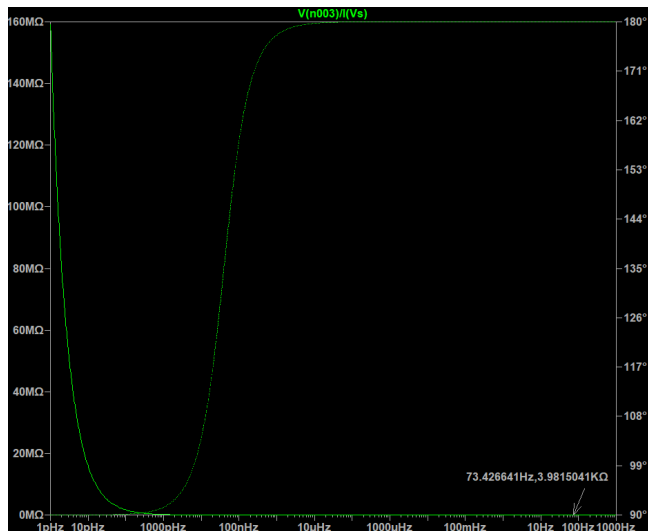
For $R_{o,t}$ we get:



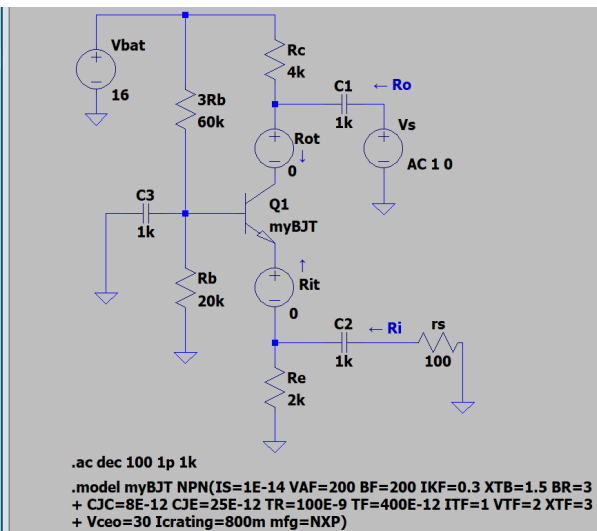
Giving $R_{ot} = 861.056 \text{ k } \Omega$



Lastly for R_o :

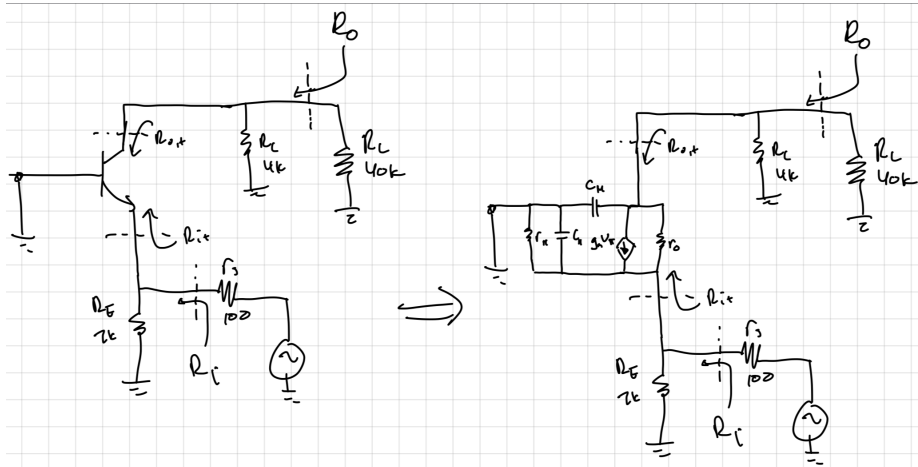


Giving $R_o = 3981.50 \text{ } \Omega$

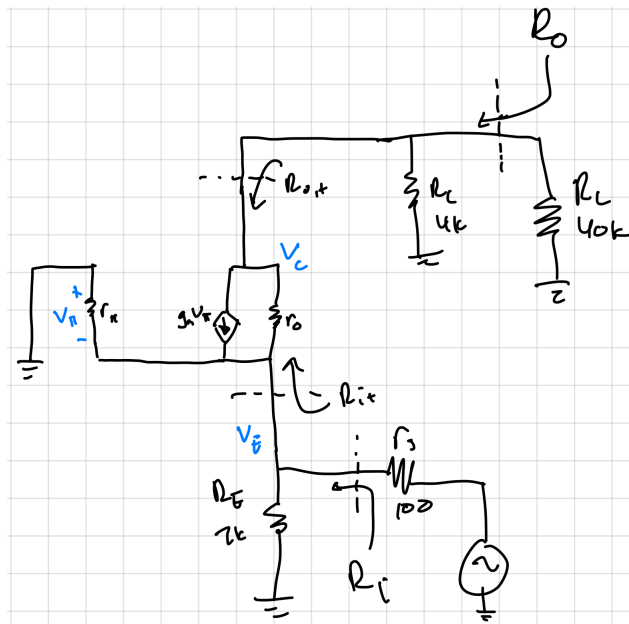


Calculation of value

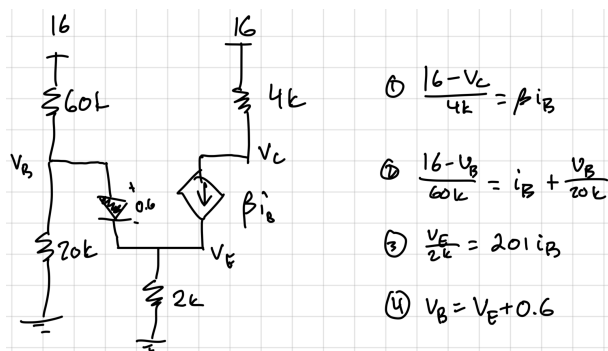
In the midband, we short the large capacitors and can transform the BJT to our small signal model, and we then short our voltages since we're only looking for the resistance:



This further simplifies to:



For the DC analysis, we see that we end up with the exact same circuit as the DC analysis in part 3, therefore we can reuse the numbers we got:



$$\begin{aligned} \textcircled{1} \quad & \frac{16 - V_C}{4k} = \beta i_B \\ \textcircled{2} \quad & \frac{16 - V_B}{60k} = i_B + \frac{V_B}{20k} \\ \textcircled{3} \quad & \frac{V_E}{2k} = 201 i_B \\ \textcircled{4} \quad & V_B = V_E + 0.6 \end{aligned}$$

vb	3.87769784173
vc	9.47721822542
ve	3.27769784173
ib	8.15347721823E-6

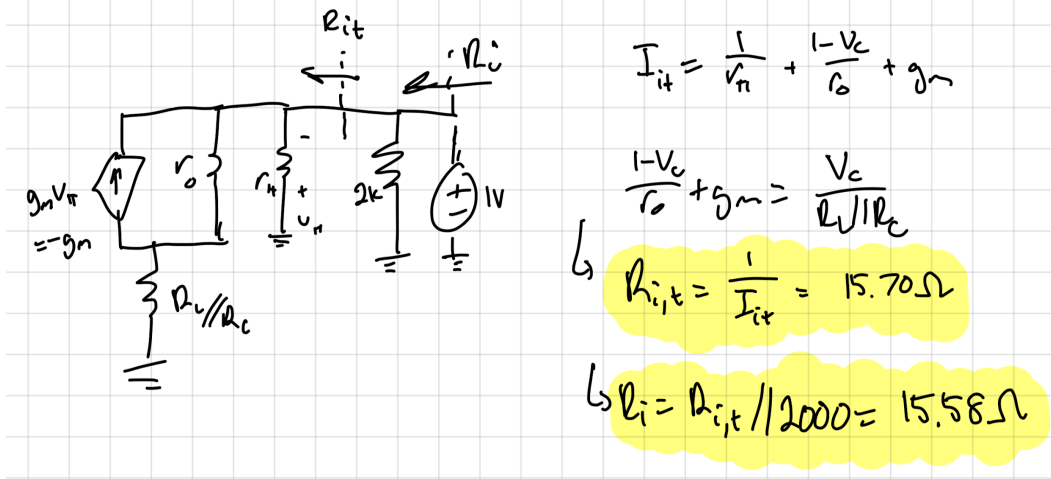
Therefore we get the same values for r_{π} and r_o :

$$r_{\pi} = \frac{V_T}{I_B} = \frac{0.025}{8.15347722E-6} = 3066.176 \Omega$$

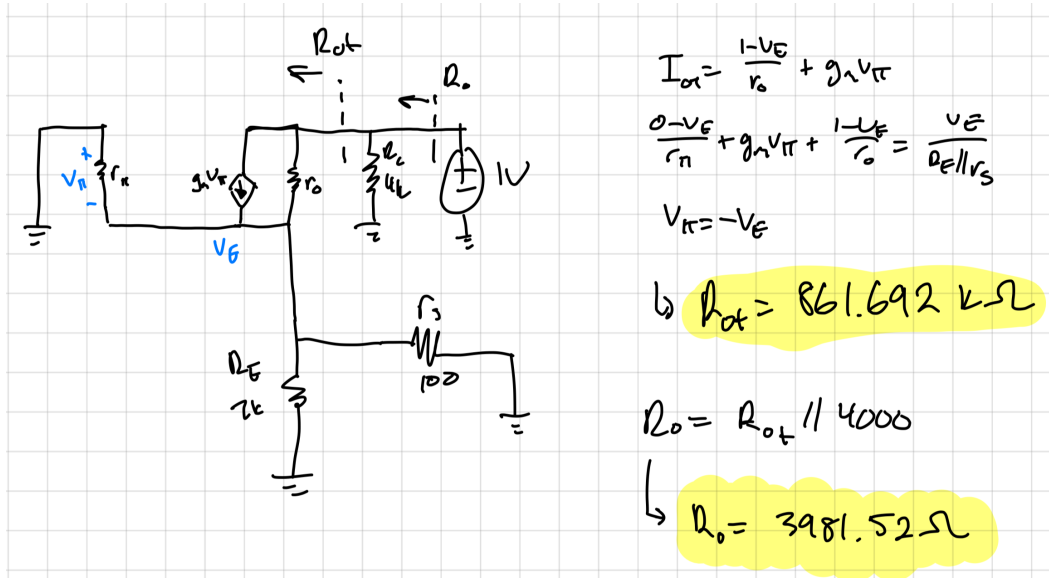
$$r_o = \frac{V_A}{I_C} = \frac{200}{1.63069544E-3} = 122.647 \text{ k}\Omega$$

Because of the dependent voltage source, we can't just zero the voltages. Instead, we can look at the circuit with respect to the resistance we are trying to find, add a 1V voltage source, and solve for the reciprocal of the current through the wire we expect the resistor to be at

For $R_{i,t}$ and R_i (we can solve them together since we can easily see $R_i = R_{i,t} \parallel R_E$):



For $R_{o,t}$ and R_o (combined since we can easily derive $R_o = R_{o,t} \parallel R_C$):



4A Summary

Resistor	Simulated Value (Ω)	Calculated Value (Ω)	Error
R_i	16.486	15.58	-5.50%
$R_{i,t}$	16.623	15.70	-5.55%
$R_{o,t}$	861056	861692	0.0739%
R_o	3981.50	3981.52	5.02E-4 %

Comparison between (3A) and (4A)

Below A table comparing all the values

Resistor	3A Simulated Value (Ω)	3A Calculated Value (Ω)	4A Simulated Values (Ω)	4A Calculated Values (Ω)
R_i	2710.30	2545.8	16.486	15.58
$R_{i,t}$	3308.01	3066.2	16.623	15.70
$R_{o,t}$	3879.30	3873.66	861056	861692
R_o	128559	122850	3981.50	3981.52

We observe that 4A has substantially lower input resistances and a lower R_o . In return $R_{o,t}$ for 4A is substantially higher than that in 3A, but overall the resistances in 4A are smaller.